

FREDERICK KWAKU RICHTER

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SUMMARY OF QUALIFICATIONS

- Embedded systems and robotics engineer with C++/Python experience developing firmware, telemetry, sensor integration, and peer-to-peer communication layers for autonomous micro-UAV systems.
- Hands-on with STM32-based flight controllers, RTOS-based ArduPilot/ChibiOS firmware, UART/SPI/I2C peripherals, MAVLink message framing, companion computers, and TCP/UDP telemetry bridges.
- Google Summer of Code contributor developing AP_SwarmMesh, a decentralized C++ peer-to-peer mesh state exchange framework for the ArduPilot open-source autopilot ecosystem.
- Researching decentralized sensor fusion and collaborative state estimation methods for GPS-denied micro-UAV swarms, connecting onboard localization, peer-to-peer communication, and resilient autonomy.
- Comfortable working across hardware, firmware, Linux/Python tooling, and operator facing validation workflows for constrained robotic systems.

EMBEDDED SYSTEMS & ROBOTICS EXPERIENCE

Google Summer of Code Contributor

05/26 - Present

ArduPilot, Remote

- Developing *AP_SwarmMesh*, a C++ ArduPilot communication middleware library for decentralized MAVLink peer state exchange over a dedicated peer-to-peer serial interface.
- Designing an object-oriented mesh architecture with stream builders, frame/deframe logic, bounded peer state tables, duplicate suppression, TTL/staleness handling, forwarding rules, and logging hooks.
- Implementing a compact peer routing header that wraps standard MAVLink frames without changing payload definitions, preserving compatibility with existing ArduPilot/MAVLink abstractions.
- Validating the protocol using UDP SITL backends with multi-vehicle relay scenarios to quantify packet correctness metrics, latency, loss, and CPU/memory usage before hardware deployment.
- Collaborating with the upstream ArduPilot community through code reviews, technical design discussions, documentation, and incremental integration with existing firmware architecture.

Defensive Drone Swarming Graduate Researcher

01/25 - Present

University of Ottawa, Ottawa ON

- Developed *ArduSwarm*, an open-source micro-UAV testbed for decentralized swarm robotics, integrating ArduPilot firmware, Bitcraze Crazyflie hardware, onboard sensing, and peer-to-peer communication.
- Wrote and integrated C++ ArduPilot HAL drivers for PMW3901 optical flow over SPI and VL53L1X Time-of-Flight ranging over I2C to enable infrastructure-free indoor localization.
- Integrated a multi-processor architecture by routing MAVLink over UART between an STM32 flight controller, ESP32/GAP8 companion computer, WiFi telemetry link, and high-level swarm command interface.
- Built a Python TCP/UDP bridge that connects to the external companion computer over TCP, unwraps payloads, forwards telemetry to PC over UDP, and receives commands through a paired UDP port.
- Implemented thread-safe bidirectional forwarding with socket locks, non-blocking reads, buffer parsing, struct binary packing, graceful shutdown, and defensive error handling.
- Validated embedded flight software through indoor experiments, achieving 0.353 m 3D localization RMSE against ground truth and 5 ms / 333 ms communication and control latencies respectively.

TECHNICAL SKILLS

- **Languages:** C++, Python, C, Lua, MATLAB, Bash
- **Embedded / MCU:** STM32F405, nRF51, GAP8 RISC-V, ESP32-C3, RPi Zero, ChibiOS/RTOS, UART, SPI, I2C, ADC, DMA, timers, GPIO, DShot/BlueJay
- **Robotics / Autonomy:** ArduPilot, MAVLink, EKF sensor fusion, optical flow, ToF ranging, indoor localization
- **Networking / Middleware:** TCP/UDP sockets, serial protocols, P2P mesh, packet fragmentation/reassembly
- **Software Engineering:** Object-oriented design, bounded buffers, state machines, binary protocols, logging, hardware-in-loop validation, code reviews, technical documentation

ADDITIONAL ENGINEERING EXPERIENCE

HVAC Systems Design Engineer

09/23 - 12/23

HTS Engineering, Vaughan ON

- Designed 63 commercial and residential refrigerant heating/cooling systems, balancing performance, cost, code compliance, and constructability.
- Developed an automated Excel pipeline tracker to improve project visibility, reporting, and team coordination across active design work.

Mechanical Building Design Engineer

01/23 - 04/23

Arup, Toronto ON

- Produced heating/cooling plant schematics and lab plumbing models for the University of Toronto Lash Miller Chemistry Building Expansion, supporting an on time Detailed Design package for costing.
- Collaborated with the civil team to draft a new underground trench and chamber service-connection proposal for the University of Toronto Rotman building, reducing spatial requirements by approximately 20%.

Asset Management Engineer

05/20 - 12/21

NRX AssetHub, Markham ON

- Restructured SAP asset data, digital parts books, and maintenance manuals for industrial clients, reducing document search time by at least 25% on Shell offshore datasets.
- Led three co-op students to build SAP asset structures and maintenance work instructions for Diageo Virgin Islands facilities, delivering two completed systems to the client.

EDUCATION

Mechanical Engineering (Thesis) – MASc

01/25 - Present

University of Ottawa, Ottawa, ON

- **Awards:** Research Assistantship (2025), Special Merit Scholarship (2025), Admission Scholarship (2025).
- **Technical Electives:** Machine Learning for Control Systems, Mechatronics, Industrial Organization.

Mechanical Engineering (Co-op) – BASc

09/19 - 06/24

University of Waterloo, Waterloo, ON

- **Awards:** Graduated with distinction (top 15%), Capstone Best Prototype award (2024), Dupont-Peppler Award for Black & Indigenous students (2023), President's Scholarship (2020).

SELECTED PROJECTS, PUBLICATIONS & AWARDS

K. Richter, N. Mailhot, M. Pirani, and D. Spinello, "**ArduSwarm: A Self-Contained Micro-Aerial Vehicle Testbed for Experimental Swarm Flight Control**," Proceedings of the CSME-CFDSC-CSR International Congress, Vancouver, BC, 2026. **2nd Best Written Paper Award**.

Raspberry Pi Zero 2 W Audio Streamer – Personal project

Designed and built a compact headless Linux audio receiver that provides Spotify Connect playback over Wi-Fi with automatic Bluetooth A2DP fallback, routing audio through an I2S DAC HAT to powered speakers or an analog stereo receiver.

PackerPro Automated seed packaging machine – 5th year Capstone; Best Prototype Award.

Sponsored by Richters Herbs to develop a low-cost, fully automated seed packaging machine which inputs bulk seeds (over 1000 varieties), measures and dispenses seeds, and outputs filled packets; showcased an improved system capable of producing 200 packets in any given production run at a rate of 15 packets per minute with a standard deviation of 0.963, increasing production by 375%; recipient of the Best Prototype award out of 42 total competing teams.

LEADERSHIP ACTIVITIES

Clubs: Senior Leader, National Society of Black Engineers (NSBE), University of Waterloo chapter; supported mentorship and events for Black students in STEM.

Electronics: Designed a custom SSD enclosure and optimized its thermal performance using CFD analysis.

Sports: Competed in alpine ski racing; recipient of MVP alpine skiing award.